

Shaocheng (Leno) Wu

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EDUCATION

University of Illinois, Urbana-Champaign		
Program: PhD in Mechanical Science and Engineering	Champaign, IL	Starting Aug 2026
Advisor: Callan Luetkemeyer, PhD		
Purdue University, West Lafayette		
Program: BS in Mechanical Engineering & Minor in Global Engineering Studies	West Lafayette, IN	Dec 2025
GPA: 3.64; Dean's List/ Semester Honors All Semester		
Study Aboard - Nanyang Technological University (NTU)	Singapore	May 2024
Program: Study Abroad Program at NTU School of Mechanical and Aerospace Engineering		
Study Abroad - Management Center Innsbruck (MCI)	Innsbruck, Austria	May 2025
Program: Short-term Study Abroad at MCI Sports and Medical Devices Research Institute		

RESEARCH INTEREST

Computational and Experimental Biomechanics, Soft-Tissue Mechanics, Image-Based Mechanical Modeling, Sports Injury Prevention, Non-Invasive Diagnostic Imaging

RESEARCH EXPERIENCE

Tendon Health Monitoring system - Purdue ME Capstone		West Lafayette, IN	August 2025 – December 2025
Project Manager / Team Lead			
- Won the second place of Malott Innovation Competition within Purdue School of Mechanical Engineering - Led the development of an integrated ankle brace–ultrasound system for real-time monitoring of Achilles tendon fatigues condition and potential injury risks. - Designed a wearable ultrasound interface that improves signal transmission, ensures user safety, and conforms to brace geometry through CAD modeling, FEA evaluation. - Designed Experiments for teammates to conduct pre-exercise and post exercise tests, using their pre and post exercise ultrasound-based acoustic responses as indicators of tendon fatigue and injury risk. - Implemented MATLAB signal processing and a custom GUI to track signal changes and provide continuous, real-time data acquisition tailored to different users.			
Purdue Ray. W. Herrick Labs - Meta-Acoustics Lab		West Lafayette, IN	May 2024 - August 2025
Summer Undergraduate Research Fellow (2024,2025) / Undergraduate Research Assistant			
Advisor: Dr. Junfei Li at Purdue School of Mechanical Engineering & Ray. W. Herrick Laboratory			
- Collaborated with Dr. Junfei Li, multiple principal investigators, and graduate students focus on designing marine environmental noise attenuation structure for offshore wind farm energy construction noise. - Applied acoustic waveguide theory using COMSOL Multiphysics 6.2 and MATLAB to design a passive noise mitigation panel effective in underwater environments, achieving significant noise reduction (>25 dB) below 1000Hz to protect marine ecosystems from anthropogenic noise pollution. - Utilized COMSOL Multiphysics 6.2 to model, analyze and optimize specialized engineered acoustic structures using polymer-based materials (PDMS, super-elastic polymers, acrylic) with air-entrained structures, creating low-impedance boundaries for noise mitigation. - Designed and fabricated a lab-scale sound source using SolidWorks to simulate pile-driving noise within a controlled water tank environment. - Fabricated and Tested lab-scale prototypes, conducting sound transmission experiments using MATLAB, evaluating Sound Pressure Difference with and without the acoustic barrier to validate its noise reduction performance across (100-20kHz). - Served as co-first author on a peer-reviewed publication in <i>Physical Review Applied</i>, independently responsible for project conceptualization, methodology, experimental validation, and full manuscript preparation including writing and editing. - Publication: Zhang, Y., Wu, S. (Co-First Author), Li, R., Oh, H., Liu, Y., & Li, J. (2026). Foldable acoustic metamaterial for underwater broadband low-frequency noise mitigation. <i>Physical Review Applied</i>, 25(2), 024029. https://doi.org/10.1103/3j5q-66c3			

Curriculum Vitae

Purdue Additive and Intelligent Manufacturing (AIM) Lab

West Lafayette, IN

Oct 2022 - Dec 2023

Undergraduate Research Assistant - Lead by Professor Huachao Mao, Dr. Yujie Shan from Purdue Polytechnic Institute

- Collaborated with graduate students to perform Process Optimization for Vat Polymerization based Additive Manufacturing
- Utilized SolidWorks to design and model a low-cost method combining LCD 3D printing with resin drop-coating, experimentally characterizing diffraction patterns and optical path differences.
- Contributed as **third author** to the design and validation of diffraction gratings fabricated by two-photon polymerization, analyzing precision limitations and secondary grating effects.
- Publication: J. Hua, Y. Shan, S. Wu, and H. Mao, "Three-Dimensional Printed Diffraction Gratings Drop Coated by Different Resins and Their Mechanism," Journal of Manufacturing Science and Engineering, vol. 146, no. 1, 011001, 2023. DOI: 10.1115/1.4063137

INDUSTRY EXPERIENCE

Aptiv Shanghai Test Center

Shanghai, China

Jun 2023 - Aug 2023

Hardware Product Engineer Intern

- Successfully completed the project (Modal Analysis for New Energy Vehicle Power Distribution Box).
- Applied materials with higher damping performance to improve over 20% durability and stability of Power Distribution Box according to experimental data acquired from LabView Software, perfectly met the requirements from European Clients
- Created more than thirty 3D Models of Power Distribution Box and optimized the Model Design using SolidWorks and Abaqus to further improve performance and reliability.
- Evaluated performance of four different types of specialty glues used inside Box by applying glue to a specific fixture and using Universal Testing Machines (MTS&ZWICK) to determine when it reached a critical fracture.

OTHER EXPERIENCE

2025 Purdue Summer Undergraduate Symposium Research Talks (July 2025)

2025 Purdue Spring Undergraduate Research Conference Poster (April 2025)

2024 Purdue Summer Undergraduate Symposium Research Talks (July 2024)

2024 Noise-Control Conference Classic Paper Presentation (June 2024)

LEADERSHIP EXPERIENCE

Global Engineering Ambassador – Affiliated with Purdue Global Engineering Programs and Partnerships

ME 290(Global Engineering Seminar-Introductory Course for New ME students) Peer Mentor

Purdue Mechanical Engineering Alumni Mentorship Program

Purdue Mechanical Engineering Tutor for ME 308 Fluid Mechanics

AWARDS AND HONORS

Purdue Mechanical Engineering Senior Design Malott Innovation Competition Second Place (December 2025)

Purdue Mechanical Engineering (GRIT) Recognition for Multidisciplinary Experience (December 2025)

Purdue Engineering Student Council (PESC) Gold Scholarship (February 2025)

Purdue Mechanical Engineering General Scholarship (June 2024)

Purdue Mechanical Engineering H. William Bottomley Research Scholarship (January 2025)

Office of Engineering Undergraduate Research Scholarship (August 2024)

Purdue Summer 2024 Reward (May 2024)

Purdue Summer 2025 Reward (May 2025)

Undergraduate Project Award (Issued by Institute of Noise Control Engineering - USA, June 2024,2025)

Hallberg Foundation Award (Issued by Institute of Noise Control Engineering - USA, June 2024)

COURSEWORK

Biomechanics, Thermodynamics, Control and Measurement I II, Engineering Acoustics, Fluid Mechanics, Heat Mass Transfer, Product Design & Development, Computer-Aided Design, Statics, Dynamics, Advanced Engineering Mathematics; Non-Destructive Testing, Solid Mechanics, Additive Manufacturing in Industry 4.0; Human-Factors Engineering Studies; Globalization Engineering Education Studies, Introductory to Material Science

ADDITIONAL SKILLS

- **Technical Skills:** COMSOL, Abaqus, MATLAB, Python, Fusion 360, SolidWorks, Siemens-NX, GD&T, LabView, Microsoft 365
- **Language:** English (Proficient), Mandarin (Native)